

substantially different pricing schedules or models.¹⁷⁶⁴ The vast majority (>85%) of on-exchange trading volume executes on exchanges with maker-taker pricing models and with baseline access fees near the cap and rebates slightly lower than the access fee cap.¹⁷⁶⁵ The few exchanges which deviate from this pricing style do not execute a large proportion of trading volume.¹⁷⁶⁶ Additionally, smaller or newer exchanges (which do not belong to one of the three large exchange families) have adopted similar pricing schedules and thus do not seem to be competing for order flow by differentiating their pricing schedule. For example, MEMX, the exchange with the most market share not affiliated with one of the three large exchange families, adopted a similar maker-taker pricing schedule to what is prevalent in the market. Additionally, IEX, a formerly flat-fee exchange, has recently switched to a maker-taker pricing model, citing the need to incentivize liquidity provision.¹⁷⁶⁷ This is in line with the discussion in section VII.C.2, which explains how the structure of markets today incentivize the adoption of maker-taker pricing where access fees are set at or near the access fee cap in order to fund large maker rebates as a means of attracting competitively priced quotes, which in turn increase the trading volume executed on the exchange. Lowering the access fee cap does not change this dynamic, and so the

¹⁷⁶⁴ As explained in section VII.C.2, exchanges can differentiate themselves by offering different fee schedules – e.g., inverted, flat fee, or maker-taker with numerous price strata. Reducing the access fee cap can reduce the variation in rebates and fees across venues by narrowing the viable range for fees and rebates thereby making the different exchange price schedules more similar. However, many price schedules are already quite similar despite the 30 mil access fee cap allowing for a greater degree of differentiation. For instance, the data reported in Table 4 does not show that there is currently a large degree of variation in the highest fees charged, particularly among maker-taker exchanges which dominate the market. Additionally inverted exchange fees are all set close to the access fee cap.

¹⁷⁶⁵ See Table 5.

¹⁷⁶⁶ Id.

¹⁷⁶⁷ Self-Regulatory Organizations; Investors Exchange LLC; Notice of Filing and Immediate Effectiveness of Proposed rule Change Pursuant to IEX Rule 15.110 to Amend IEX's Fee Schedule, Securities Exchange Act Release No. 98063 (Aug. 4, 2023), 88 FR 54373 (Aug. 10, 2023).

Commission expects that exchanges will continue to set fees and rebates at or near the lowered access fee cap.

When stating that a lower access fee cap would limit competition by restricting differentiation, one commenter pointed out that when one exchange switched from a flat rebate model to a tiered pricing model that exchange quoted at the NBBO more often.¹⁷⁶⁸ This is consistent with a tiered pricing structure discouraging order routing to competing venues. In this case switching to a tiered pricing schedule incentivized that exchange's members to not route orders to competing exchanges to collect the benefits associated with high volume tiers. More orders sent to the exchange incentivizes more aggressive quoting on the exchange leading the exchange to quote at the NBBO more often. However, the effect on tiering from the amendment's reduction in the access fee cap would be different from this example because the reduction in the access fee cap would apply to all exchanges, meaning that the effect on competition from tiering will also be diminished across all exchanges. Therefore, the effect that a lower access fee cap would have on one exchange's ability to more consistently quote at the NBBO is likely to be weaker than that stated by the commenter.

One commenter expressed the concern that the reduced access fee cap may also result in exchanges increasing the cost to access market data or sell preferential access to exchange data to some members but not to others.¹⁷⁶⁹ The adopted amendments address this concern as compared to the proposal by eliminating the proposed requirement for an access fee cap of 5 mils on some stocks. One commenter stated that while it is possible that a sufficiently low cap could

¹⁷⁶⁸ See Cboe Letter III at 8.

¹⁷⁶⁹ See RBC Letter at 4.

generate this concern, the access fee cap of 10 mils strikes the right balance.¹⁷⁷⁰ Furthermore, as explained in section VII.D.2.b, the Commission does not expect exchange transaction revenues on transactions priced greater than \$1.00 to substantially change as a 10 mil access fee cap is expected to be high enough that exchanges can continue to realize their current net capture rates for these transactions.

Reducing the access fee cap may also impact competition between broker-dealers who are exchange members, to the extent that a lower access fee cap diminishes the marginal benefit of qualifying for a pricing tier with lower fees or higher rebates over another tier with less preferential terms. Different transaction pricing tiers, particularly volume-based pricing tiers providing more favorable fees and/or rebates to exchange members who execute higher relative order volume, introduce a competitive wedge between those exchange members who qualify for the better tiers and those who do not. Another commenter stated that high access fees disproportionately affect smaller firms and investors, and lowering the access fee cap would promote a more competitive and diverse market landscape.¹⁷⁷¹ The commenter stated that exchanges employ pricing tiers to extract rents from smaller exchange members which are then split between the exchange and their high tier members, and lowering the access fee cap would limit the extent to which this can occur.¹⁷⁷² Under the assumption that a reduction in access fees would be accompanied by a reduction in transaction rebates, one possible effect of reducing the access fee cap would be to diminish the relative differences in the fees charged and rebates

¹⁷⁷⁰ See Verret Letter I at 8.

¹⁷⁷¹ See Verret Letter I at 9.

¹⁷⁷² Id. at 6, 7.

offered between different pricing tiers.¹⁷⁷³ As shown in Table 4 multiple exchanges have fee or rebate tiers which vary within a range that is greater than 10 mils. Therefore, lowering the access fee cap to 10 mils will necessitate that pricing tiers would have to be placed within a narrower price range. The Commission expects that as the differences between pricing tiers become less economically meaningful, the competitive wedge introduced by pricing tiers will diminish, which would make it easier for exchange members lacking scale to compete with exchange members that qualify for preferential pricing tiers (i.e., tiers with higher rebates/lower fees).

Another commenter stated that although the access fee cap would be lowered, it could still allow sufficient room for existing differences in preferential pricing to persist.¹⁷⁷⁴ The commenter used the example of a large bank and small broker both paying a 30 mils rebate with the bank receiving a 32 mils rebate and the broker receiving 24 mils. Under a 10 mils access fee cap, the exchange could offer a 12 mils rebate to the bank and 4 mils to the broker. In that case the differential between the bank and the broker would remain the same despite the reduced access fee cap. In the example provided by the commenter, the exchanges could continue to offer a fully funded rebate to some exchange members, which is 8 mils greater than offered to other exchange members, because that 8 mil differential would be allowed under a 10 mil access fee cap. The Commission acknowledges that exchanges would be able to continue to offer differentiated pricing; however, a lower access fee cap would reduce the extent to which pricing tiers can differ and limit the aggregate fees available to the exchange to redistribute among its

¹⁷⁷³ According to Table 4 the differences in pricing tiers for many exchanges exceed 10 mils therefore if the number of pricing tiers does not decrease, by necessity, the average difference between the tiers would diminish.

¹⁷⁷⁴ See Healthy Markets Letter I at 23-24.

members in the form of rebates. It is more difficult for an exchange to fund high rebates, particularly those greater than the fee cap, under a lower access fee cap.

**c. Acceleration of the MDI Rules, Addition of Information
About Best Odd-Lot Orders, Fees and Rebates Determinable
at Time of Execution**

Accelerating the inclusion of odd-lot information in the NMS data, along with the implementation of the MDI Rules round lot definition, might lead to increased competition between exchanges and ATSs and OTC market makers, including wholesalers. NMS stocks priced greater than \$250.00 are expected to benefit sooner from a tighter NBBO, thereby increasing the competitiveness of the best displayed protected quotes. Greater visibility of more competitively priced odd-lot orders inside the NBBO could increase the competitive position of exchanges and ATSs and attract greater order flow. This effect will be temporary, only lasting until the full implementation of the MDI Rules. After the full implementation of the MDI Rules, the effect on competition is accounted for in the MDI Adopting Release and is not ascribed to these amendments.

Making exchange fees and rebates determinable at the time of execution will enable the customers of broker-dealers to better discuss transaction fees and rebates with their broker-dealers, and potentially request data on the exchange fees incurred by an order,¹⁷⁷⁵ which will increase competition between broker-dealers along this dimension, leading to better order

¹⁷⁷⁵ For example, some brokers allow customers to direct an order to a particular exchange. By making the fees and rebates determinable at execution, the broker may be better able to inform the customer of the net transaction price of a prospective directed order.

execution and lower costs.¹⁷⁷⁶ In particular, while there is currently no requirement to either pass on the fees and rebates to customers, or account for fees and rebates when assessing execution quality, there may be competitive pressure to do so as a result of the amendments because a competing broker-dealer will be able to include fees and rebates in its transaction cost analysis, or simply pass them through to the customer. One commenter stated that the requirements for exchange pricing under this rule change will be “even more anti-competitive” than the current practice, because this would mean that “smaller brokers can’t attract new flows based on modelling of what such flows will do to their rates upon arrival.”¹⁷⁷⁷ The Commission disagrees with the statement that this rule change will make the market for offering executing broker services more anti-competitive. As described in section VII.D.3, this rule change allows brokers to determine their fees and rebates at execution and thereby eliminates the need for forecasting future market outcomes in order to anticipate the fee that will be incurred by an order. To the extent such forecasting is more difficult for small brokers, the rule change will make it easier for small brokers to compete.

Including odd-lot information in the exclusive SIPs and providing the best odd-lot order information will enhance competition among broker-dealers. Making the best odd-lot order information accessible through the exclusive SIPs will facilitate better analysis of a broker-

¹⁷⁷⁶ Under the baseline it would be difficult in many cases for a broker-dealer to allocate specific rebates received or fees paid to one customer’s trade because the fees or rebates in a given month are based, in many instances, on that broker-dealer’s total trading volume across all customer accounts, see section VII.C.2.b. However, if the fees and rebates are determinable at the time of execution the broker-dealer could feasibly track a specific fee or rebate to a specific trade, making it possible for a customer to receive such information.

¹⁷⁷⁷ See Danny Mulson Letter.

dealer's execution quality than is available with just NBBO data.¹⁷⁷⁸ Thus, it could be easier for some customers to monitor the performance of their broker-dealers.¹⁷⁷⁹

Accelerating the inclusion of odd-lot data into the exclusive SIPs will increase competition among data providers of odd-lot information prior to the full implementation of the MDI Rules, though it will do so less than envisioned in the MDI release for the period until the MDI Rules are fully implemented. Specifically, under the implementation schedule in the MDI Rules, adding odd-lot information to core data was to occur during the parallel operation period. Adding odd-lot information to the current exclusive SIPs will enable the exclusive SIPs to compete directly with the exchanges' proprietary data products for use in visual display settings. Without this change, the only means to get odd-lot information is to subscribe to multiple proprietary data feeds. This will change when odd-lots are a part of SIP data.

Unlike the data provided by the competing consolidators, the current exclusive SIPs are not fast enough for use in certain trading.¹⁷⁸⁰ Thus, the competition for odd-lot data will be limited to odd-lot information used in visual display settings. To the extent that some market participants subscribe to proprietary data for use in visual display settings, the introduction of odd-lot information to the exclusive SIPs will provide competition to this segment of the market and reduce the prices of odd-lot information provided by the proprietary data feeds. However, the Commission does not believe that this market is very large. Currently, for most display

¹⁷⁷⁸ See supra note 1621, for a discussion on the incentives that institutional traders have to monitor all aspects of transaction costs.

¹⁷⁷⁹ It is possible that some institutional traders have access to proprietary data feeds that provide the ability to benchmark trades against odd-lot orders. Or, they could contract with specialized firms that have access to the data and provide transaction cost analysis.

¹⁷⁸⁰ See MDI Rules for a discussion of the SIPs' higher latency relative the proprietary feeds offered by exchanges. In particular, footnote 26 on page 18599 summarizes commenters' views on the disadvantages of using SIP data instead of proprietary feeds.

settings, market participants use SIP data or one of the top of book data products offered by one of the three highest volume exchange groups; it is unclear to what extent market participants subscribe to proprietary data with odd-lot information for use in visual display settings.

With respect to competition for top-of-book (TOB) data, the exclusive SIPs face competition from exchanges' TOB data products. As discussed in the MDI adoption,¹⁷⁸¹ these proprietary products are typically less expensive and contain less content—being derived from a single exchange or exchange family—than the exclusive SIPs. If the exclusive SIPs charge more for data on account of the increased costs associated with disseminating odd-lot information, then this may provide a competitive advantage to providers of proprietary TOB products.

Requiring the exclusive SIPs to disseminate the accelerated odd-lot information until the exclusive SIPs are retired will guarantee that the odd-lot information will be disseminated.¹⁷⁸² However, this requirement may also affect competition among competing consolidators once the MDI Rule is fully implemented. On the one hand, these new requirements on the SIPs could reduce competition among competing consolidators and therefore reduce the expected benefits of the MDI Rules. This reduction in competition could occur because the amendments may increase the competitive advantage of exclusive SIPs relative to non-SIP competing consolidators because the SIPs will have established a market for odd-lot information before having to face competition. That is, the SIPs will have time to acquire customers for odd-lot information before other competing consolidators can enter. These customers may then face costs should they

¹⁷⁸¹ See MDI Adopting Release, supra note 10 at 18603.

¹⁷⁸² See supra section VII.D.4.c for additional discussion. While the amendments require the exclusive SIPs to distribute odd-lot data, the MDI Rules do not require the competing consolidators to disseminate odd-lot data. However, the MDI Adopting Release anticipated that at least one competing consolidator will do so because there would be demand for the data.

switch to a non-SIP competing consolidator; these switching costs may dissuade entry by non-SIP competing consolidators and thereby lower competition.¹⁷⁸³

On the other hand, the Commission is uncertain whether the SIPs will become competing consolidators.¹⁷⁸⁴ The amendments' requirement for SIPs to disseminate odd-lot information reduces the incremental costs that the SIPs would need to bear in order to become competing consolidators. Therefore, these amendments make it more likely that the SIPs will register as competing consolidators, which would improve competition relative to a scenario in which they do not compete.

Further, non-SIP competing consolidators will still have an opportunity to compete for significant market share. As discussed above, SIPs face latency disadvantages relative to exchanges.¹⁷⁸⁵ If competing consolidators can offer a lower latency product, then they can capture a part of the market that the amendments will not affect—those customers who will use odd-lot information in ways other than visual display.¹⁷⁸⁶ Likewise, competing consolidators can offer depth-of-book data under the MDI Rules, which the SIPs are not required to disseminate under these amendments. If these markets are significantly bigger than the odd-lot visual display market, the competitive advantage of the exclusive SIPs will be less likely to dissuade entry, and non-SIP competing consolidators could have sufficient incentive to enter the market.¹⁷⁸⁷

¹⁷⁸³ See MDI Adopting Release, supra note 10, for further discussion of how competing consolidators have higher barriers to entry than exclusive SIPs, such as in the form of compliance costs associated with Regulation SCI.

¹⁷⁸⁴ See supra note 1782.

¹⁷⁸⁵ See supra note 1780.

¹⁷⁸⁶ See discussion around note 1780, supra explaining that the SIPs' latency disadvantage makes their data useful for visual display.

¹⁷⁸⁷ In the MDI Adopting Release, the Commission anticipated that both exchanges operating exclusive SIPs would have strong incentives to enter the competing consolidator market. See MDI Adopting Release, supra note 10, at 18761.

3. Capital Formation

The Commission expects that the amendments will promote capital formation. First, the combined effect of the amendments will be to increase liquidity generally, which will increase incentives to trade and therefore price efficiency. Price efficiency in turn promotes capital formation. The Commission also expects that the alleviation of tick constraints and the lower access fee cap will work together and separately to lead to displayed prices that are more reflective of supply and demand for the underlying securities, also promoting capital formation.

One commenter stated that a narrower tick could increase volatility and decrease liquidity which could discourage companies from going public.¹⁷⁸⁸ As discussed in section VII.D.1, stocks receiving the \$0.005 tick on average will not experience harmful liquidity effects. On the contrary, as discussed in section VII.D.1, the expectation is that on average liquidity will improve for stocks with narrow quoted spreads that receive the tick size reduction. Additionally, as discussed in section VII.D.1, the narrower tick will not result in increased volatility.¹⁷⁸⁹ Consequently, even if there was a link between liquidity and volatility, and the decision to go public, those channels aren't expected to be affected in the manner mentioned by the commenter. Further, the link between tick sizes and IPOs is not clearly defined in existing research.¹⁷⁹⁰

¹⁷⁸⁸ See RBC Letter at 3.

¹⁷⁸⁹ See *supra* notes 1209 and 1210 and surrounding text for a discussion of tick sizes and volatility.

¹⁷⁹⁰ Research on this topic is exceptionally difficult. As stated in the report Assessment of the Plan to Implement a Tick Size Program, “There are myriad factors influencing companies’ decisions about whether to go public or remain private — and, if an IPO is desired, in which country to list shares. These include the availability of capital outside the public equity market, the regulatory burdens placed on public companies, market conditions, broader macroeconomic trends and differences in economic conditions between countries globally. Additionally, broader historical context may reveal certain periods of strong IPO issuance, particularly during times of high speculative activity in markets, as anomalous and unsustainable.” See Securities and Exchange Commission, Assessment of the Plan to Implement a Tick Size Pilot Program (Jul.3, 2018), available at <https://www.sec.gov/files/TICK%20PILOT%20ASSESSMENT%20FINAL%20Aug%202.pdf> (last accessed Feb. 6, 2024).